

AVESH SHAH

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Website: <https://www.aves-shah.com>

Core Skills

- Controls & Systems: Estimation (Kalman/RLS), LQR/LQI, torque management, vehicle dynamics, Torque path control strategies, Model Predictive control
- Technical Tools: MATLAB/Simulink/Stateflow, Python, C++, ROS2, Docker, LabVIEW,
- Testing & Validation: DSpace, INCA, ETAS, MotoHawk, National Instruments, Carla
- Process & Tools: AUTOSAR, ASPICE, DOORS, Polarion, Git/Azure , JIRA, ISO26262, CAN , LIN
- Management skills: Project Planning, Negotiate timeline, Change management

Professional Experience

Magna International - Troy, MI

Function Owner | September 2025- present

- Leading the development of an MPC-based vehicle trajectory tracking and roll-avoidance strategy using differential braking and bounded steering adjustments; utilizing CasADi expressions with acados for real-time optimization."
- Collaborate with ADAS and validation teams to review test results, debug issues, and update software baselines.
- Supporting development of a Gateway module consisting of Vehicle control and a connectivity system integrating Wifi and GNSS.

Senior Software Engineer-LMDD | Apr 2024 – September 2025

- Performed CODEC configuration activities for speakers and Microphones. Developed ROS2/Docker interface with CODEC to playback Alerts and Advertisements.
- Developed ML based Algorithm using Tensorflow-Lite to detect Emergency vehicle approaching using Yamnet as a reference. Re-trained the model and validated it over 50000 audio files. Used ROS2 and Docker to integrate it with custom built S32G based Connectivity Board
- Calibrated IMU from scratch removing static Bias and used Mahogany and Kalman filters to calculate Roll and Pitch of a 3 wheeled Delivery vehicle.
- Developed and validated estimation and control algorithms for vehicle state (Vx, Vy, heading, pose) using slip aware sensor fusion (IMU, GPS, wheel speeds).
- Integrated Ublox GNSS-based positioning using NTRIP corrections into ROS2 and Docker pipelines for achieving cm-level accuracy and transmitted the parameters over CAN.
- Developed a Mass Estimation algorithm using RLS filter using 2 Lambda functions , achieving <15% maximum error.

Dana Incorporated - Novi, MI

Technical Lead | Jan 2023 – Apr 2024

- Led a team of 6 Engineers for software feature development for BEV transmission systems under AUTOSAR and ASPICE frameworks.
- Contributed to project planning, task definition, and coordination with program management and OEM partners.
- Designed and implemented control logic for torque management, lubrication, and thermal handling features.
- Conducted internal technical reviews and supported feature validation and calibration activities.
- Assisted in defining software test coverage and resolving integration and validation issues.

FEV North America - Auburn Hills, MI

Sr. Controls Engineer | Oct 2018 - Jan 2023

- Developed torque path control algorithms including creep, dual-path shift energy management, e-coasting, One Pedal Drive, Brake Blending and cost-based torque split optimization for hybrid and BEV programs.
- Collaborated and lead workshops with suppliers to align on requirements, timeline, and performance validation criteria.
- Designed engine start/idle control profiling using LQR for faster actuation with minimal delays.

Hybrid Design Services Inc. - Troy, MI

Controls Engineer | Dec 2016 - Oct 2018

- Built prototype BMS with SOC estimation using EKF, power limit calculation, and contactor control with weld check logic.
- Developed a BDU unit serving as a watchdog system that can override the client's BMS system to secure the Battery pack.
- Implemented supervisory monitoring controller to monitor and log faults via OBD-II port of a Dumpster truck and send logged data over FTP to secure server.
- Performed third-party BMS calibration and HV battery pack testing for fuse current, isolation monitoring, contactor weld check, and cell balancing.

General Motors (Contract) - Pontiac, MI

R&D Test Systems Engineer | Aug 2014 - Dec 2016

- Developed Simulink/LabVIEW controls for gasoline and diesel research engines.
- Built FPGA/DSPACE test systems for prototype ECU validation and CAN communication.
- Performed test cell commissioning by integrating Rapid Control Prototype ECUs with test environments and calibrating sensors such as MAF, cam phaser, crank encoders, thermocouples, and solenoids.

Education

M.S. - Automotive Engineering (GATE Fellow) - Clemson University | GPA 3.57

B.Tech- Mechanical Engineering - National Institute of Technology, Jaipur | GPA 8.37

Certification - Applied Machine learning